

OLA

Data Analysis Project Report

SQL • Python • Power BI

1. Project Overview

This project analyzes ride-booking data of, a ride-hailing platform, covering 103,024 bookings recorded between July 1–31, 2024. The goal is to understand booking success and cancellation patterns, revenue and payment behavior, vehicle-type performance, and customer/driver ratings — using SQL for structured business analysis and Power BI for interactive visualization — in order to surface insights that can guide operational and business decisions.

The analysis is organized into three parts:

- SQL Analysis – ten targeted business questions answered with SQL queries and their results
- Power BI Dashboard – a five-page interactive dashboard covering overall performance, vehicle types, revenue, cancellations, and ratings
- Key Insights & Recommendations – takeaways and suggested next steps based on the combined findings

2. Dataset Summary

Metric	Value
Total Bookings	103,024
Date Range	01-Jul-2024 to 31-Jul-2024
Columns	20
Vehicle Types	Prime Sedan, Prime SUV, Prime Plus, Mini, Auto, Bike, eBike
Payment Methods	Cash, UPI, Credit Card, Debit Card
Booking Statuses	Success, Canceled by Customer, Canceled by Driver, Driver Not Found

Key Columns

- Identifiers & Time: Date, Time, Booking_ID, Customer_ID
- Trip Details: Booking_Status, Vehicle_Type, Pickup_Location, Drop_Location, Ride_Distance
- Service Times: V_TAT (vehicle turnaround time), C_TAT (customer turnaround time)
- Cancellations: Canceled_Rides_by_Customer, Canceled_Rides_by_Driver
- Incomplete Rides: Incomplete_Rides, Incomplete_Rides_Reason
- Financials: Booking_Value, Payment_Method
- Ratings: Driver_Ratings, Customer_Rating

3. SQL Analysis

Ten business questions were answered using SQL queries against the bookings table. Each question below shows the query used and the resulting answer.

1. Retrieve all successful bookings

```
SELECT * FROM bookings WHERE Booking_Status = 'Success';
```

63,967 bookings (62.1% of all 103,024 bookings) completed successfully.

2. Find the average ride distance for each vehicle type

```
SELECT Vehicle_Type, AVG(Ride_Distance) AS avg_distance FROM bookings GROUP BY Vehicle_Type;
```

Vehicle Type	Avg. Distance (km) – Successful Rides
eBike	25.15
Prime Sedan	25.01
Prime Plus	25.03
Mini	24.98
Bike	24.93
Prime SUV	24.88
Auto	10.04

Auto trips are noticeably shorter than every other vehicle type — consistent with Autos typically being used for short, local hops rather than longer cross-town rides.

3. Get the total number of cancelled rides by customers

```
SELECT COUNT(*) FROM bookings WHERE Booking_Status = 'Canceled by Customer';
```

10,499 rides were cancelled by customers (10.2% of all bookings).

4. List the top 5 customers who booked the highest number of rides

```
SELECT Customer_ID, COUNT(Booking_ID) AS total_rides FROM bookings GROUP BY Customer_ID ORDER BY total_rides DESC LIMIT 5;
```

Customer ID	Total Rides
CID954071	5

Customer ID	Total Rides
CID836942	4
CID309168	4
CID952434	4
CID199017	4

Note: this is a different ranking from “Top 5 Customers by Total Booking Value” shown in the Power BI dashboard — the highest-frequency riders are not the same customers as the highest-spending ones.

5. Get the number of rides cancelled by drivers due to personal and car-related issues

```
SELECT COUNT(*) FROM bookings WHERE Canceled_Rides_by_Driver = 'Personal & Car related issue';
```

6,542 rides were cancelled by drivers specifically for personal or car-related issues — the single largest reason within driver-side cancellations (35.5% of the 18,434 driver-cancelled rides).

6. Find the maximum and minimum driver ratings for Prime Sedan bookings

```
SELECT MAX(Driver_Ratings) AS max_rating, MIN(Driver_Ratings) AS min_rating FROM bookings WHERE Vehicle_Type = 'Prime Sedan';
```

Maximum rating: 5.0 | Minimum rating: 3.0. Despite this spread, the average Prime Sedan driver rating sits at 3.99, indicating most ratings cluster tightly near the top end.

7. Retrieve all rides where payment was made using UPI

```
SELECT * FROM bookings WHERE Payment_Method = 'UPI';
```

25,881 rides were paid via UPI — the second most common payment method behind Cash.

8. Find the average customer rating per vehicle type

```
SELECT Vehicle_Type, AVG(Customer_Rating) AS avg_customer_rating FROM bookings GROUP BY Vehicle_Type;
```

Vehicle Type	Avg. Customer Rating
Prime Plus	4.01

Vehicle Type	Avg. Customer Rating
Auto	4.00
Mini	4.00
Prime SUV	4.00
Prime Sedan	4.00
Bike	3.99
eBike	3.99

Customer ratings are essentially flat across vehicle types (4.00 ± 0.01) — vehicle type alone does not appear to meaningfully drive customer satisfaction in this dataset.

9. Calculate the total booking value of rides completed successfully

```
SELECT SUM(Booking_Value) AS total_successful_value FROM bookings WHERE
Booking_Status = 'Success';
```

₹35,080,467 in total booking value was generated from successful rides.

10. List all incomplete rides along with the reason

```
SELECT Booking_ID, Incomplete_Rides_Reason FROM bookings WHERE Incomplete_Rides =
'Yes';
```

Incomplete Reason	Count
Customer Demand	1,601
Vehicle Breakdown	1,591
Other Issue	734

3,926 rides were marked incomplete in total. Customer Demand and Vehicle Breakdown are nearly tied as the leading causes, together accounting for over 81% of incomplete rides.

4. Power BI Dashboard

An interactive five-page dashboard was built in Power BI to explore the dataset visually, with a date-range filter applied across views (01-Jul-2024 to 31-Jul-2024).

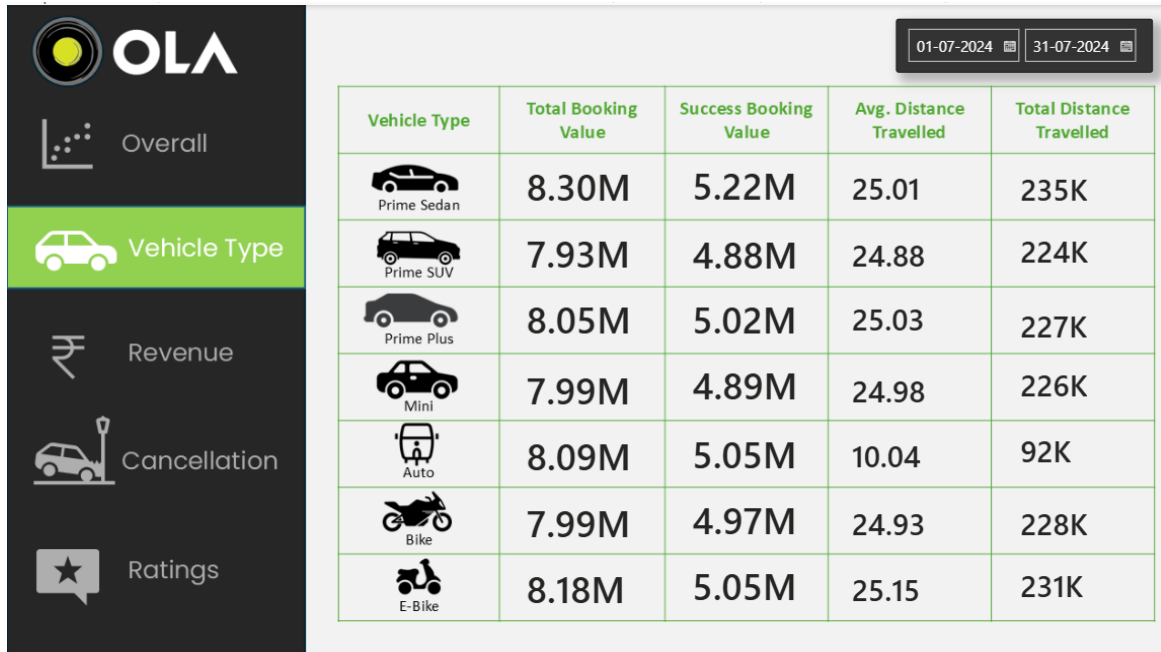
4.1 Overall



Total Bookings: 103,024 | Total Booking Value: ₹35M

Bookings hold a fairly steady daily volume (roughly 3,200–3,450 per day) with a sharp drop on the final day of the period. Of all bookings, 62.1% completed successfully, while cancellations by drivers (17.9%) outnumber cancellations by customers (10.2%), and 9.8% failed because no driver was found.

4.2 Vehicle Type



Booking value is fairly evenly distributed across vehicle types (₹7.9M–8.3M each), and each type converts a little under two-thirds of its bookings to successful value. Average distance for successful rides is consistent at ~25 km across Sedan, SUV, Plus, Mini, Bike, and eBike — except Auto, which averages just 10.04 km, reflecting its role in shorter local trips.

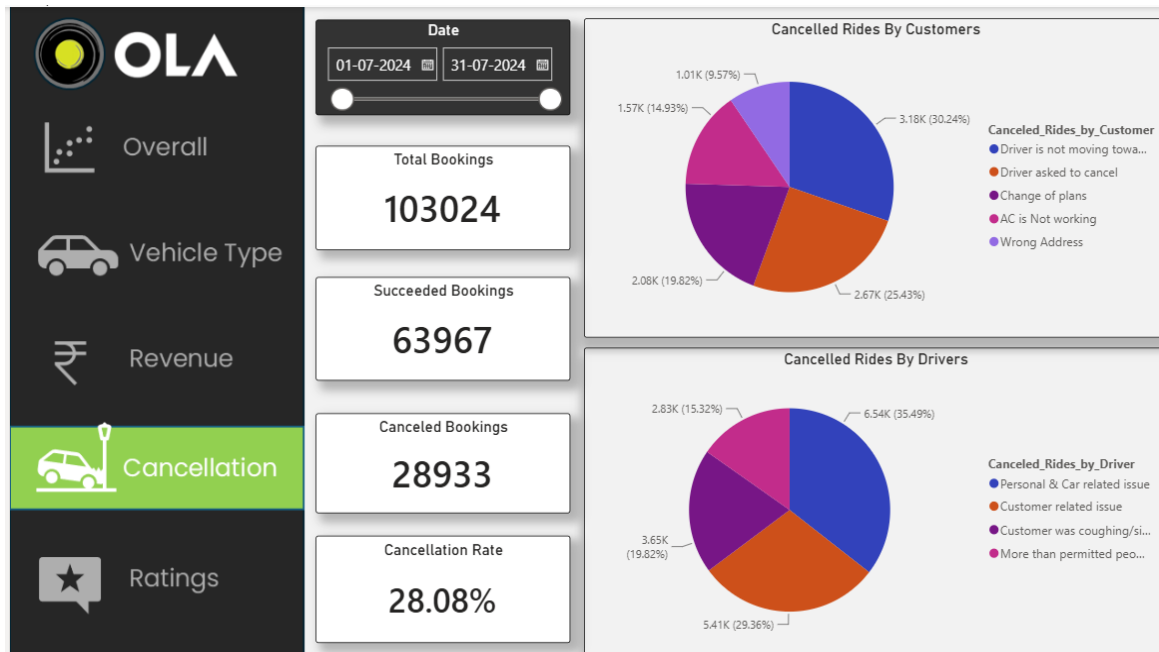
4.3 Revenue



Cash is the dominant payment method by revenue, followed by UPI; Credit Card and Debit Card make up a small share of total revenue. Daily revenue is stable throughout the month (roughly ₹45K–55K per day). The top 5 customers by total booking value each spent between ₹6,000

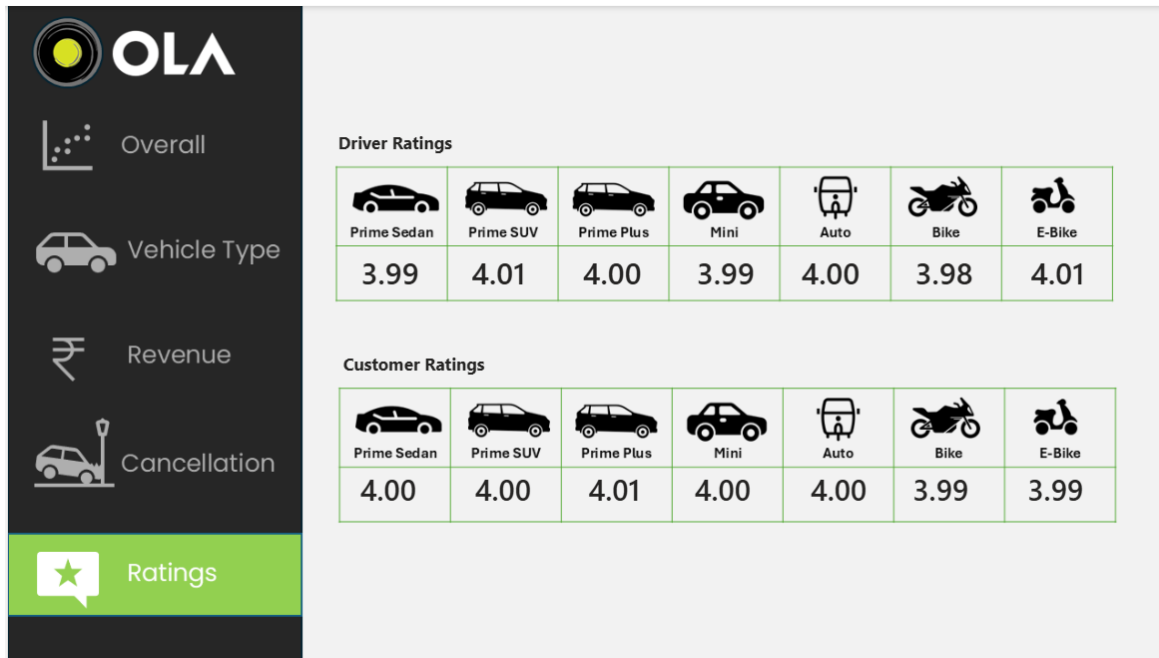
and ₹8,025 over the period, a different group than the top 5 by ride count identified in the SQL analysis.

4.4 Cancellation



Of 103,024 total bookings, 28,933 were cancelled, for a 28.08% overall cancellation rate. On the customer side, “Driver is not moving towards pickup” (30.2%) and “Driver asked to cancel” (25.4%) are the leading causes. On the driver side, “Personal & car related issue” (35.5%) and “Customer related issue” (29.4%) dominate — suggesting cancellations are driven roughly equally by driver-side logistics issues and friction between customers and drivers.

4.5 Ratings



Both driver and customer ratings are tightly clustered between 3.98 and 4.01 across every vehicle type, with no vehicle type standing out as notably better or worse rated. This mirrors the SQL finding in Question 8 — vehicle type has little to no effect on satisfaction in this dataset.

5. Key Insights

- Cancellations are a major leakage point: 28.08% of all bookings are cancelled, with driver-initiated cancellations (17.9% of all bookings) outweighing customer-initiated ones (10.2%).
- Driver-side cancellations are split almost evenly between logistics (“Personal & car related issue,” 35.5%) and people-related friction (“Customer related issue,” 29.4%) — these likely need different fixes.
- Auto stands apart from every other vehicle type — shorter average trips (~10 km vs. ~25 km) consistent with short, local-hop usage rather than cross-town rides.
- Ride frequency and spend don’t align: the top 5 customers by number of rides are a completely different set from the top 5 by total booking value, meaning loyalty and high-value spending are two separate customer behaviors worth targeting differently.
- Ratings are uniformly high and flat (~4.0) across all vehicle types for both drivers and customers — satisfaction isn’t a differentiator between vehicle types in this dataset.
- Cash and UPI dominate payments, together covering the large majority of revenue, while card payments remain a small share — worth watching if OLA wants to shift toward digital payment adoption.
- Incomplete rides (3,926 total) are driven almost equally by Customer Demand and Vehicle Breakdown, pointing to both fleet maintenance and customer-experience factors as causes worth investigating further.